

Alex Liu

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EDUCATION

University of Florida, *B.S. Computer Science, Minor in Geography*

GPA:3.80/4.0 | Gainesville, FL

• **Award:** Dean's List | Expected Graduation: May 2028

• **Relevant Coursework:** Computer Organization, Linear Algebra, Data Structures and Algorithms, Programming Fundamentals 2, Discrete Structures, Calculus 3

SKILLS

- **Languages:** Python, C++, C, Java, JavaScript, TypeScript, SQL, RISC-V Assembly, Bash, HTML/CSS, GO, C#
- **Frameworks & Libraries:** React.js, Node.js, FastAPI, Flask, TailwindCSS, SQLAlchemy, Pandas, NumPy
- **Developer Tools:** Git, GitHub, Docker, GitHub Actions (CI/CD), Linux/Unix, VS Code, Postman, Vite
- **Platforms & Databases:** PostgreSQL, Supabase, Vercel, Railway, Microservices, REST APIs, JWT

EXPERIENCE

Instructor, *Logic Lab*

Jun 2022 – Jul 2022 | Gainesville, FL

- Led and instructed a class of **30+** students in math and chess, designing lesson with 4 other team members.

PROJECTS

Geography Dashboard - AI-Assisted Geospatial Analytics Platform,

Nov 2025

React, Leaflet, FastAPI, Vite, TypeScript, PostgreSQL, Docker, Gemini AI 

- Engineered a **full-stack** geospatial platform using a **microservices** architecture to separate high-load AI inference from core analytics, achieving **distributed deployment** via Docker to ensure independent service scaling and fault isolation.
- Systematised the production workflow by establishing a **CI/CD pipeline** with GitHub Actions, automating build validation & deployment to Railway, reducing configuration errors & ensured environment synchronisation.
- Integrated a cost-aware **AI** insight layer using Google Gemini, implementing strict database-level rate limiting to optimize API usage and enforce responsible resource consumption across multi-user sessions.
- Developed deterministic spatial algorithms using **FastAPI** and **SQLAlchemy**, transforming raw CSV/GeoJSON datasets into interactive, city-scale visualizations & quantitative urban insights.

UF Health SmartScribe, *HTML, CSS, JavaScript, Supabase, Gemini AI* 

Nov 2025

- Engineered a real-time **full-stack** medical documentation prototype within **24 hours**, integrating Deepgram and Gemini API to transcribe consultations and auto-generate structured SOAP notes.
- Architected a secure backend pipeline using **Supabase** to manage patient records, automating clinical workflows to reduce documentation time by **60%**.
- Implemented a speaker-aware AI extraction engine processing **60+ words/minute** with **sub-5-second** latency, prioritizing immediate physician utility over administrative overhead

Grade Track, *Flask, React.js, SQLAlchemy, PostgreSQL, Docker* 

Oct 2025

- Engineered a **full-stack** analytics dashboard that visualizes student performance trends across courses using Flask APIs, React, and Recharts to integrate real-time data aggregation and visual analytics to demonstrate
- Orchestrated deployment in Docker, reducing **85%** setup time ensuring consistent cross-environment builds.

Ability Bridge, *Python, OpenCV, MediaPipe, PyAutoGUI* 

Sep 2025

- Constructed a hands-free accessibility interface using **OpenCV**, processing **14,000+** facial landmarks/second to enable precise cursor control for users with limited mobility.
- Optimized signal processing algorithms (exponential smoothing) to achieve **30 FPS** with **<100ms** latency, minimizing false activations across diverse lighting conditions.

LEADERSHIPS & ACTIVITIES

UF Society of Asian Scientists & Engineers, *SASE intern*

Aug 2024 – Present | Gainesville, FL

- Planned a professional event with **100+** members dedicating **30+** hours, for **150+** attendees in SASE